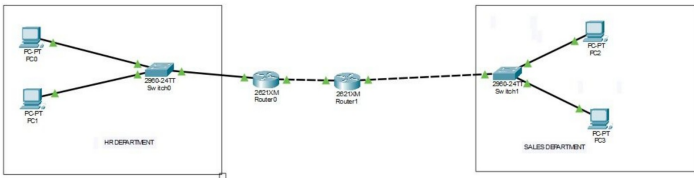


Inter-Department Network Lab - DHCP and Static Routing

★ Overview

This lab simulates two company departments — **HR** and **Sales** — connected across two routers over a WAN link. Each department’s router hands out IP addresses dynamically via **DHCP**, and the two networks reach each other using a **static route**.

🏗️ Topology



Network Topology

- **HR Department:** PC0, PC1 → Switch0 → Router0 (Fa0/0)
- **Sales Department:** PC2, PC3 → Switch1 → Router1 (Fa0/0)
- **WAN Link:** Router0 (Fa0/1) ↔ Router1 (Fa0/1)

🌐 IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Assignment
Router0	Fa0/0	50.0.0.1	255.0.0.0 (/8)	Static (DHCP gateway for HR)
Router0	Fa0/1	172.16.0.1	255.255.0.0 (/16)	Static (WAN link)
Router1	Fa0/0	192.168.2.1	255.255.255.0 (/24)	Static (DHCP gateway for Sales)
Router1	Fa0/1	172.16.0.2	255.255.0.0 (/16)	Static (WAN link)
PC0	Fa0	50.0.0.52	255.0.0.0	DHCP
PC1	Fa0	50.0.0.51	255.0.0.0	DHCP
PC2	Fa0	192.168.2.51	255.255.255.0	DHCP

⚙️ DHCP Configuration

Router0 — Pool “HR” - Network: 50.0.0.0 /8 - Excluded address: 50.0.0.1 (reserved for the gateway) - Default gateway: 50.0.0.1

Router1 — Pool “Sales” - Network: 192.168.2.0 /24 - Excluded address: 192.168.2.1 (reserved for the gateway) - Default gateway: 192.168.2.1

Sample DHCP bindings on Router1:

IP Address	Client / MAC Address	Type
192.168.2.51	0001.9745.90EB	Automatic
192.168.2.52	000A.41E3.9196	Automatic

🏠 Routing Configuration

Router0 uses a static route to reach the Sales LAN through Router1:

```
ip route 192.168.2.0 255.255.255.0 172.16.0.2
```

Router0’s routing table confirms this:

```
C 50.0.0.0/8 is directly connected, FastEthernet0/0
C 172.16.0.0/16 is directly connected, FastEthernet0/1
S 192.168.2.0/24 [1/0] via 172.16.0.2
```

(For full connectivity, Router1 would similarly need a static route back to the HR network: `ip route 50.0.0.0 255.0.0.0 172.16.0.1`.)

✓ Verification Steps

1. Ran `ipconfig` on PC0-PC3 to confirm each host received a valid IP address, subnet mask, and default gateway via DHCP.
2. Ran `show ip int brief` on both routers to confirm interfaces were up/up with correct IP addressing.
3. Ran `show ip route` on Router0 to confirm the static route to the Sales subnet was installed.
4. Ran `show ip dhcp pool` and `show ip dhcp binding` to confirm DHCP scopes and active leases.

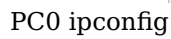
🎯 Key Learning Objectives

- Configuring DHCP pools on a Cisco router (excluded addresses, default gateway, network scope)
 - Verifying DHCP client assignment from the PC command line
 - Configuring and verifying static inter-network routing
 - Reading and interpreting `show ip route`, `show ip int brief`, and `show ip dhcp pool` output
-

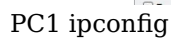
📷 Appendix — Screenshots

HR Department

PC0 — `ipconfig` output



PC1 — ipconfig output



Router0 – CLI Verification

```

Router0
-----
Physical Config CLI Attributes
IOS Command Line Interface

Router0#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 50.0.0.1        YES manual up          up
FastEthernet0/1 172.16.0.1      YES manual up          up

Router0#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C 50.0.0.0/8 is directly connected, FastEthernet0/0
C 172.16.0.0/16 is directly connected, FastEthernet0/1
S 192.168.2.0/24 [1/0] via 172.16.0.2

Router0#sh ip dhcp pool

Pool HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 16777214
Leased addresses : 2
Excluded addresses : 1
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
50.0.0.0.1        50.0.0.1 - 50.255.255.254 2 / 1 / 16777214

Router0#sh ip dhcp pool

Pool HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 16777214
Leased addresses : 2
Excluded addresses : 1
Pending event : none

```

Router0 CLI - interfaces, routing table, DHCP pool

show ip int brief, show ip route, and show ip dhcp pool output on Router0, confirming interface addressing, the static route to 192.168.2.0/24, and the active HR DHCP pool.

```

Router0
-----
Physical Config CLI Attributes
IOS Command Line Interface

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
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Router0#sh ip dhcp pool

Pool HR :
Utilization mark (high/low) : 100 / 0
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Total addresses : 16777214
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Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
50.0.0.1          50.0.0.1 - 50.255.255.254 2 / 1 / 16777214

Router0#sh ip dhcp pool

Pool HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 16777214
Leased addresses : 2
Excluded addresses : 1
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
50.0.0.1          50.0.0.1 - 50.255.255.254 2 / 1 / 16777214

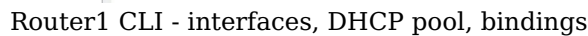
Router0#

```

Router0 CLI - DHCP pool continued

Continued show ip dhcp pool output on Router0.

Router1 — CLI Verification

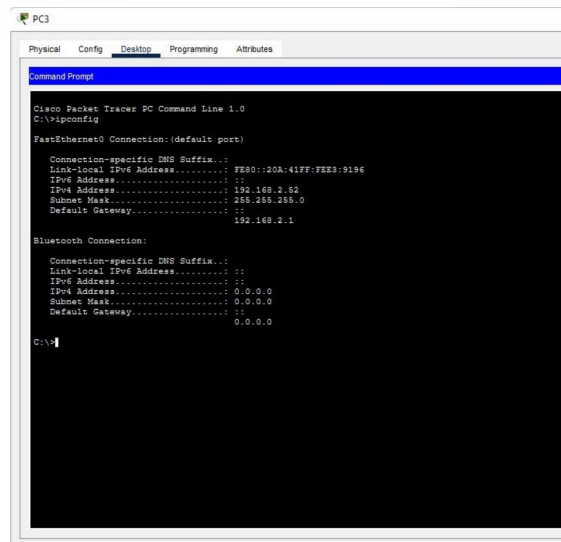


Sales Department

[illegible]

PC2 received IP address 192.168.2.51, subnet mask 255.255.255.0, and default gateway 192.168.2.1 via DHCP.

PC3 — ipconfig output



```
PC3
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::20A:41FF:FEE3:9196
    IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: 192.168.2.52
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.2.1

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
    Default Gateway . . . . .: 0.0.0.0

C:\>
```

PC3 ipconfig

PC3 received IP address 192.168.2.52, subnet mask 255.255.255.0, and default gateway 192.168.2.1 via DHCP.

Lab built and tested in Cisco Packet Tracer as part of CCNA studies.